Al-Balqa Applied University

جامعة البلقاء التطبيقية

18/30

Midterm Exam

Course Title	Control Systems	Course Code	ELE0334
Semester	First	Academic Year	2025/2026
Exam Date	1/12/2025	Day	Wednesday
Time	14:30-15:20		

Student Name [redacted] Student ID No. [redacted] Section د. خوالده

12:30 - 1:30

GENERAL INSTRUCTIONS

1. Students are not allowed to leave the exam room during the first 30 minutes of the exam.
2. Late students are not permitted to complete their exams after the first 30 minutes unless authorized by the proctor (no students left the exam session), knowing that no additional time will be granted.
3. Students must use a blue or black pen. Pencils are not allowed.
4. Consumption of any food or beverage is prohibited except in medical cases.
5. All electronic devices (e.g., cell phones) are prohibited.
6. All cell phones must be turned off and placed on the invigilator's desk in the exam room.
7. All notes and reading materials must be placed on the invigilator's desk in the exam room.
8. Students are not permitted to speak to or ask questions of their classmates. If a student has a question, he or she must raise his or her hand to be assisted by the proctor.
9. Write the full information required on this page before commencing your exam.
10. Students are not allowed to leave the room once the exam commences for any reason unless it is absolutely necessary.
11. Reported student cheating cases, misbehavior, or other student misconduct on examination sessions will be reported to the Investigations Committee as per internal bylaws.

I have understood and abide with the exam policy. **Student's signature:** [redacted]

Question No.	Question Type	CILOs assessed	Marks Weight	Scored Marks	Marker signature	Student Score
1	Multiple choice	1-3	18(66%)	20		
2	Drawing and Solution	4	7(23.33%)	10		
3						
4						
5						
6						
Total			30(100%)			

Moderation Review			
Comments			
Marks Reviewed by		Signature & Date	

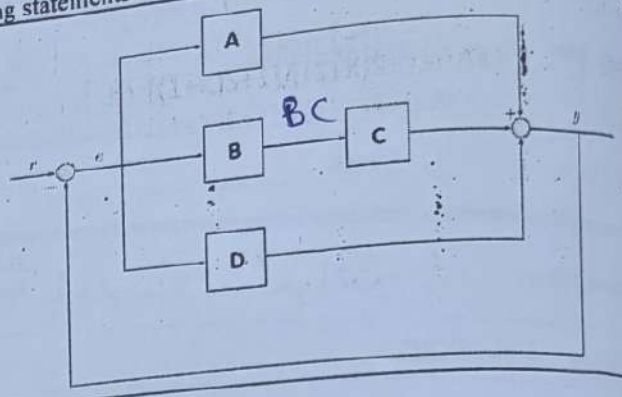
Question No. 1 (20 points)
Please select the most accurate answer for each of the following statements:

1

1. The transfer Function from Fig.1 is:

a. $A+BC+D$
b. $\frac{A+BC+D}{1+BC}$
c. $\frac{A+BC+D}{1+A+BC+D}$
d. $\frac{A+BC+D}{1+D}$
e. none

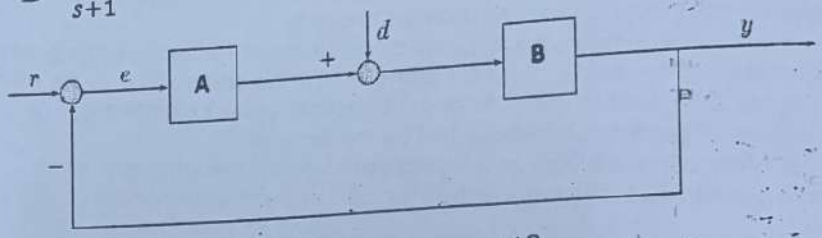
(A+BC+D)/(1+A+BC+D)



2

You are given the Below system diagram. What is the associated transfer function from disturbance to output: y/d

$A=\frac{s+1}{s+2}$ $B=\frac{1}{s+1}$



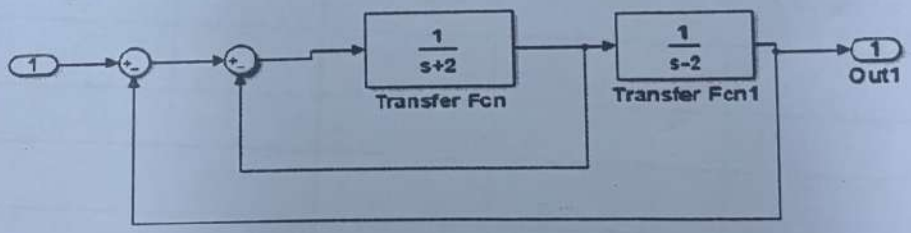
a. $\frac{1}{s(s+1)}$ b. $\frac{s+1}{s+2}$ c. $\frac{s+1}{(s+2)(s+3)}$ d. $\frac{s+2}{(s+1)(s+3)}$ e) none

3

Moving the pole of the first order system near to the origin (by making T larger) it will
A. make the system slower
B. Make the system faster
C. increase the Dc gain
D. Decrees the Dc gain
E. none

4

6. The transfer function of the following system is:



1/(s^2+5s+7)

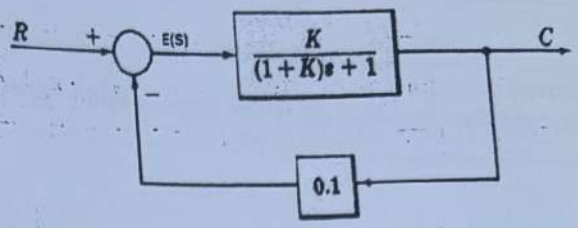
A. $\frac{1}{s^2+s-6}$ B. $\frac{1}{s^2-s-6}$ C. $\frac{1}{s^2+s-5}$ D. $\frac{1}{s^2+s+6}$ E. none

5

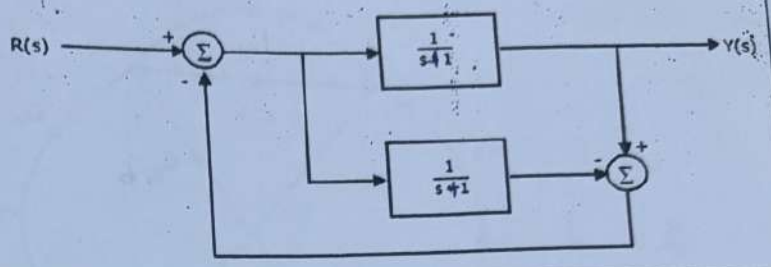
For the following first order system $\frac{C(s)}{R(s)}=\frac{10}{2s+1}$ the input is r(t)=1. At t=0 the value of c(t) is:

A. 6.32 B. 2 C. 0 D.4 E. none

6 For the block diagram shown below,
E(S)/R(S)=
a) [(S+1]/[(1+K)S+1+0.1K]
b) [(1+K)S+1]/[(1+K)S+1+0.1K]
c) [(1+K)S+1]/[(K)S+1+0.1K]
d) None



7 The transfer function of the system below is
a) $\frac{1}{s+1}$ b) $\frac{2}{s+3}$
c) $\frac{2}{s+1}$ d) None



8 for the following first order system $\frac{C(s)}{R(s)} = \frac{10}{2s+1}$ the Dc.gain is:
A.5 B.6.32 C.0 D.10 E. none

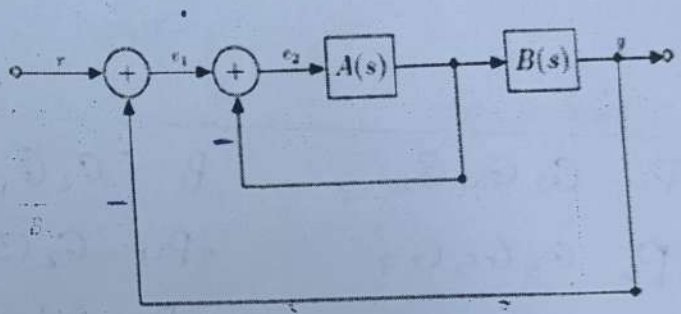
9 The transfer function relating C(s) and R(s) is

A. $\frac{B(s)}{1+A(s)+A(s)B(s)}$

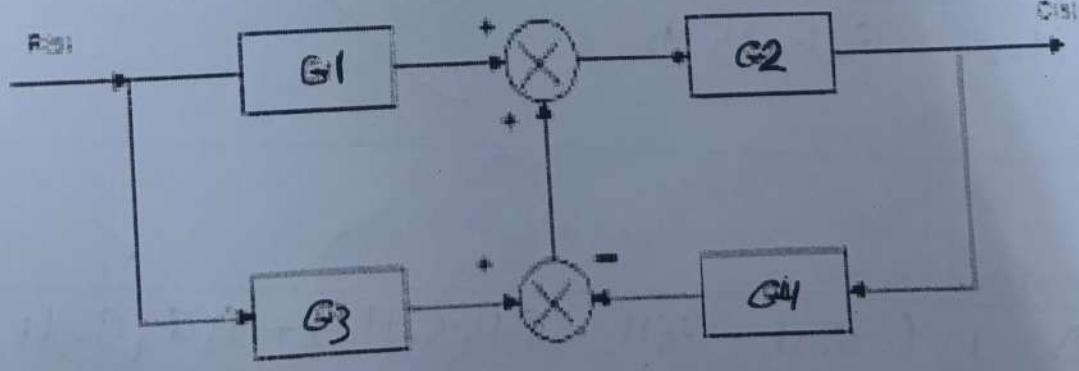
B. $\frac{A(s)B(s)}{1+A(s)+B(s)+A(s)B(s)}$

C) $\frac{A(s)B(s)}{1+A(s)+A(s)B(s)}$

D) $\frac{A(s)B(s)}{1-A(s)+A(s)B(s)}$



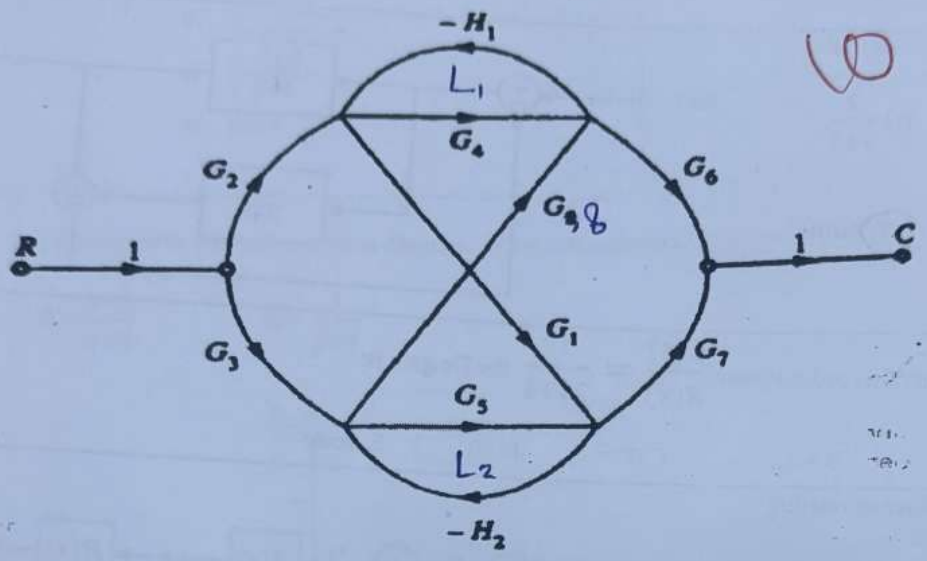
10 Consider the block diagram shown below> If the transfer function of the system is given by T(s)=G1G2+G2G3/1+X. Then X is:
a) G2G3G4 b) G2G4
c) G1G2G4 d) G3G4 e) none



1	2	3	4	5	6	7	8	9	10
a	d	A	A	E	b	d	A	C	C

8

Q2 (10 points)
Consider the control system whose signal flow graph is shown below. Determine the system transfer function using Mason's formula



10

1) Forward paths are:

$P_1 = G_2 G_4 G_6$

$P_2 = G_3 G_5 G_7$

$P_3 = G_2 G_1 H_2 G_8 G_6$

$P_4 = G_2 G_1 G_7$

$P_5 = G_3 G_8 G_6$

$P_6 = G_3 G_8 H_1 G_4 G_7$

2) Feedback loops (close loops) are:

$L_1 = -G_4 H_1$

$L_2 = -G_5 H_2$

$L_3 = +G_1 H_2 G_8 H_1$

~~$L_4 = +H_1 G_1 H_2 H_2$~~

3) Non-touch loops (2 at a time) are:

$L_1 L_2 = G_4 G_5 H_1 H_2$

4) Find the determinant Δ

$\Delta = 1 - (-G_4 H_1 - G_5 H_2 + G_1 H_2 G_8 H_1) + (G_4 G_5 H_1 H_2)$